E&ICT, IIT Roorkee



Advanced Program in

Applied Data Analytics & Generative AI

By E&ICT Academy, IIT Roorkee

6-Months 140 Hours (Projects)

About The IIT Roorkee

Indian Institute of Technology Roorkee (IIT Roorkee), one of the oldest technical institutions in Asia, was established in 1847 as the Thomason College of Civil Engineering and was later converted to an IIT in 2001.

The institute offers a wide range of academic programmes including BTech, BArch, BS, MTech, MSc, MBA, MDes, and PhD across its 23 academic departments, centres, and schools covering engineering, sciences, architecture, management, humanities, and social sciences.

Over the years, IIT Roorkee has developed state-of-the-art infrastructure, a strong research ecosystem, and a rich legacy of excellence in technical education and innovation. The institute continues to be recognized nationally and internationally for its academic rigour, research output, and contributions to industry and Society.



About The E&ICT Academy IIT Roorkee

Electronics & ICT Academy, IIT Roorkee, aims to strengthen the skills of faculty members, working professionals, and students from Engineering, Science, Commerce, and Polytechnic institutions through specialized short-term training programs.

The Academy focuses on both fundamental and advanced areas of Information Technology, Electronics & Communication, Product Design, and Manufacturing.

In addition, E&ICT Academy, IIT Roorkee, organizes customized industry-oriented training programs, faculty development programs (FDPs), and research promotion workshops to bridge the gap between academia and industry. These initiatives are designed to enhance technical competency, encourage innovation, and promote collaborative research in emerging technologies.



About The IoT Academy

The IoT Academy, established in 2017 is a fast emerging company imparting quality programs for skills training, internship and guidance in cutting edge technologies like Data Science, Machine Learning, Artificial Intelligence, Internet of Things, Embedded Systems & many more and focused on helping people develop the skills they need to thrive in the rapidly growing digital economy.

The IoT Academy has collaborated with various premier institutes e.g. E&ICT Academy, IIT-Roorkee, IIT-Guwahati and IIT-Kanpur for Advance Certification courses to take provide outcome-centric solutions to help them achieve their professional goals.

400+
Hiring Partners

30,000+
Learners

8+ Years
Of excellence

81%
Salary Hike

91%
Happy Outcomes



About The Program

Dive into the exciting world of Applied Data Analytics and Generative AI. This program helps you learn how to study data, understand it, and use it to make better decisions. You will also learn how generative AI works and how it can create new ideas and solutions. These skills can be used in many fields like healthcare, finance, retail, and technology. This course gives you the knowledge to use data and AI to solve real problems in the real world.



Mastering Data Analytics & Generative AI

- Learn the basics of data analytics. You will understand how to collect data, clean it, and show it in charts. You will also learn simple statistics.
- Generative AI helps by making new data, giving better predictions, and doing some tasks automatically.

Practical Applications

- You will work on real projects. You will explore data, build prediction models, check risks, and test different situations.
- Generative AI will help you find useful insights, work faster, and solve business problems.

Industry Trends & Real-World Impact

- You will study real examples from companies like Google, Microsoft, and IBM.
- You will learn how data analytics and generative AI are changing industries. They help companies make better decisions, give personalized services, and create new ideas.

Ethics, Governance, and Responsible AI

- You will learn about common issues like bias, data privacy, and copyright in AI.
- You will understand how to use AI in a safe and fair way by following rules and protecting data.

Live Industry-Based Projects

Industry projects are an important part of the Advanced Program in Applied Data Analytics & Generative AI. These projects let you work on real problems from real companies. You get hands-on experience in data analytics and generative AI. This helps you learn practical skills and solve tough problems in different industries.

Text-to-Image Generation

Description: Combine the power of NLP and CV to generate realistic images from textual descriptions, such as creating visual prototypes for design concepts or generating fictional landscapes.

Tools and Technologies: PyTorch, Tensor-Flow, OpenCV, Diffusion Models, GANs, Transformers.

Outcomes: Learn to preprocess text and image data, design transformer-based models, and implement diffusion models for image synthesis.





Interactive Chatbot for Healthcare

Description: Build a conversational agent using GPT-based LLMs that assists users with healthcare queries, such as symptom checks or medication reminders.

Tools and Technologies: Rasa, PyTorch, Hugging Face Transformers, Streamlit, NLP Libraries (spaCy, NLTK).

Outcomes: Gain experience fine-tuning LLMs for domain-specific tasks, handling conversational context, and deploying chatbots.

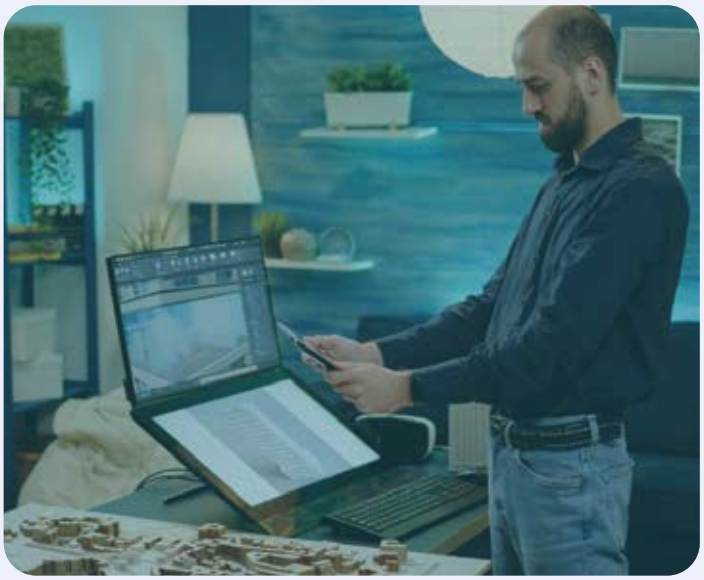
Live Industry-Based Projects

Generative Art Creation

Description: Develop a system to generate artwork using GANs or style transfer techniques. The model will create unique artistic styles or merge different styles into new creations.

Tools and Technologies: TensorFlow, PyTorch, StyleGAN, OpenCV.

Outcomes: Understand GAN architecture, apply advanced loss functions, and explore applications in digital art and design.





Predictive Maintenance with Generative Models

Description: Use Variational Autoencoders (VAEs) to detect anomalies in industrial sensor data and predict maintenance needs.

Tools and Technologies: PyTorch, TensorFlow, VAE, Pandas, NumPy.

Outcomes: Learn latent space representation, anomaly detection, and how generative AI can enhance predictive maintenance.

Live Industry-Based Projects

AI-Powered Content Generation for Marketing

Description: Create a generative AI model to assist marketing teams in generating product descriptions, blog posts, and ad copies.

Tools and Technologies: GPT, Hugging Face, PyTorch, TensorFlow.

Outcomes: Work with fine-tuning strategies for LLMs, data cleaning, and evaluating text quality using BLEU and ROUGE metrics.





Video Synthesis for Personalized Education

Description: Build a system that uses GANs to create educational video content tailored to individual learning styles and topics.

Tools and Technologies: GANs, TensorFlow, PyTorch, FFMPEG for video processing.

Outcomes: Understand video generation techniques, data preprocessing for videos, and model evaluation.

Live Industry-Based Projects

Voice Cloning and Audio Synthesis

Description: Develop a model to synthesize human-like speech or clone specific voices using generative models like WaveNet or Tacotron.

Tools and Technologies: TensorFlow, PyTorch, Librosa, WaveNet.

Outcomes: Learn to preprocess audio data, generate waveforms, and build real-time voice applications.





Smart Surveillance System Using Generative Models

Description: Detect and track anomalies or objects in video footage using generative models for data augmentation and anomaly detection.

Tools and Technologies: TensorFlow, PyTorch, OpenCV, GANs.

Outcomes: Explore applications in computer vision, object detection, and real-time surveillance.

Live Industry-Based Projects

AI-Powered Story Writing

Description: Create an LLM-based application for collaborative storytelling, where users provide input, and the model generates the next part of the story.

Tools and Technologies: GPT, Hugging Face, Streamlit, PyTorch.

Outcomes: Implement LLMs for creative writing, build interactive web apps, and handle user-driven input generation.





Climate Impact Analysis

Description: Use generative AI to analyze and visualize the effects of climate change through image and text data synthesis.

Tools and Technologies: Diffusion Models, GANs, TensorFlow, PyTorch.

Outcomes: Learn to work with multi-modal data and generate insights using generative AI.

Live Industry-Based Projects

Analyze Sales Dataset with Pivot Tables & Charts

Description: Summarize and visualize sales data to discover insights like trends, category-wise performance, and regional sales using Excel's Pivot Tables and charts.

Tools and Technologies: Microsoft Excel, Pivot Tables, Pivot Charts

Outcomes: Learn to structure raw sales data, create summaries, and visualize using pie, bar, column, and line charts for actionable business intelligence.





Customer Dataset Cleaning & Analysis

Description: Cleanse customer records to remove duplicates, fix data format issues, and fill missing values, ensuring robust analysis and reliable business decisions.

Tools and Technologies: Excel formulas, Python (pandas), SQL

Outcomes: Master systematic approaches for identifying errors, standardizing data fields, and validating customer information for downstream analysis.

Live Industry-Based Projects

Explore Sales Dataset Distribution

Description: Analyze the spread of sales figures using statistical measures and distribution plots to identify outliers and assess sales variability.

Tools/Technologies: Excel, Python (matplotlib, seaborn), Histograms, Box Plots

Outcomes: Develop the ability to compute summary statistics, generate distribution charts, and detect anomalies or patterns within sales data.





Customer Purchase Analysis with SQL

Description: Query sales and customer databases to investigate purchasing patterns, loyalty, and segmentation using relational SQL analyses.

Tools/Technologies: MySQL/SQL Server, SQL queries (JOIN, GROUP BY, aggregates)

Outcomes: Learn to extract key metrics such as total purchases, customer segments, and retention trends directly from structured datasets.

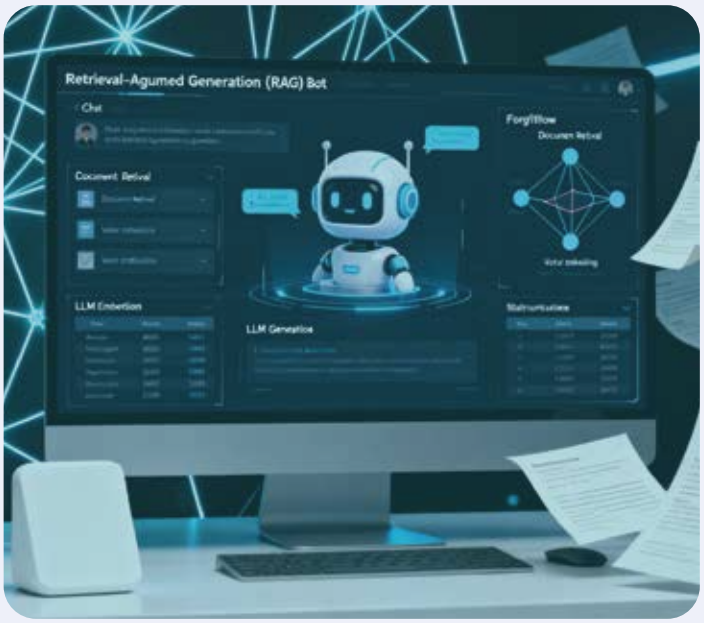
Live Industry-Based Projects

Sales Dashboard in Excel

Description: Build an interactive dashboard consolidating sales KPIs, trends, and geographic data using Excel features such as slicers and dynamic charts.

Tools/Technologies: Excel Dashboards, Pivot Tables, Slicers, Conditional Formatting

Outcomes: Gain practical skills to design, customize, and automate dashboards that help visualize and track important sales metrics.





RAG-Based Business Q&A Bot

Description: Create a Retrieval-Augmented Generation (RAG) bot that answers business questions using both structured and unstructured company data.

Tools/Technologies: Python, Transformers, OpenAI/LLMs, Retrieval pipeline (Haystack, LangChain)

Outcomes: Understand text preprocessing, transformer-based retrieval, and generation architectures to enable informed business Q&A and analytics.

Live Industry-Based Projects

Sentiment Analysis of Customer Feedback

Description: You will use NLP and machine learning to study customer comments. The system will mark each comment as positive, negative, or neutral. This helps you understand customer feelings and what they like or dislike.

Tools: Python (NLTK, spaCy, scikit-learn, BERT), TensorFlow, and libraries for charts.

Outcomes: You will clean the text, build a sentiment model, and create charts to show trends. You will also learn how to use the results to improve products or services.





Visualization of Life Expectancy and GDP Over Time

Description: You will study how life expectancy and GDP change across countries and years. You will make interactive charts to show patterns and differences.

Tools: Python (pandas, matplotlib, seaborn, Plotly), Jupyter Notebook, Excel, and public datasets.

Outcomes: You will learn how to clean data, combine datasets, and tell stories with charts. You will make useful plots like scatter, line, and animated charts.

Live Industry-Based Projects

Credit Card Customer Attrition Analysis

Description: You will analyze customer data to find out why people close their credit card accounts. You will identify at-risk customers and suggest ways to keep them.

Tools: Python (pandas, scikit-learn), SQL, charts, and models like logistic regression and decision trees.

Outcomes: You will practice data exploration, correlation analysis, and classification modeling. You will give business ideas to reduce customer loss.





Customer Segmentation Analysis

Description: You will use machine learning to group customers based on their details and buying behaviour. This helps companies offer better and personalized services.

Tools: Python (pandas, scikit-learn), SQL, charts, and clustering algorithms like K-means.

Outcomes: You will learn how to clean data, create features, and build clusters. You will understand each customer group and suggest ideas for better marketing.

Live Industry-Based Projects

Customer Growth Intelligence Dashboard

Description: Analyze customer acquisition, retention, and churn patterns using data-driven dashboards to identify growth opportunities and optimize business strategies.

Tools and Technologies: Python, Pandas, SQL, Power BI / Tableau, Machine Learning, Scikit-learn.

Outcomes: Learn to build interactive dashboards, perform customer segmentation, and generate actionable growth insights.





Startup Financial Health & Burn-Rate Radar

Description: Monitor startup financial metrics such as burn rate, runway, and profitability to assess financial sustainability and future risk.

Tools and Technologies: Python, Excel, SQL, Financial Modeling, Power BI, Machine Learning.

Outcomes: Gain experience in financial analysis, forecasting burn rate, and building investor-ready financial dashboards.

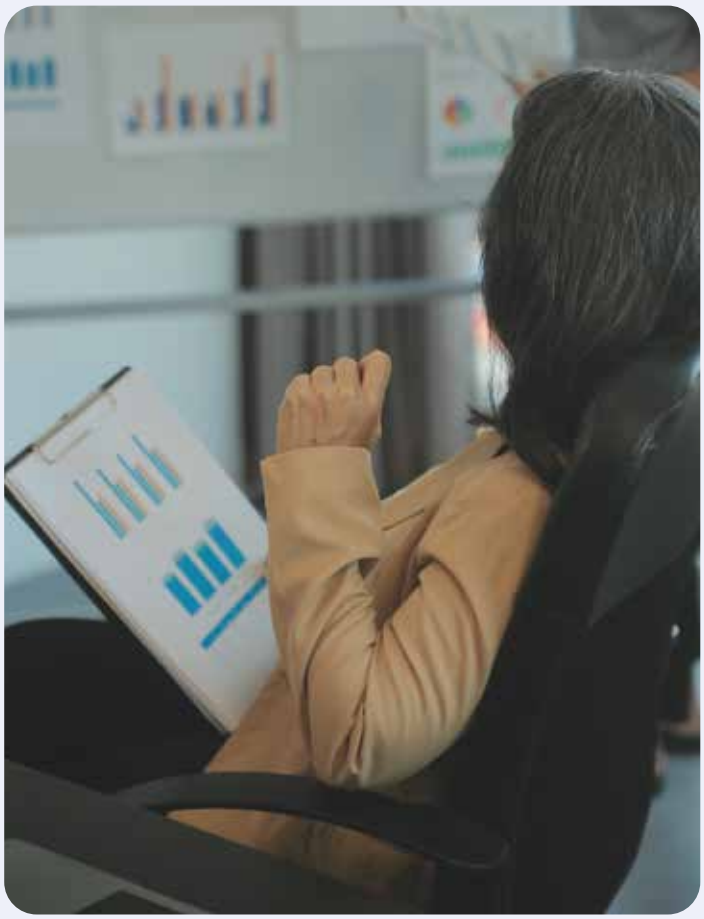
Live Industry-Based Projects

People Analytics: Silent Attrition & Workforce Signals

Description: Identify early signs of employee disengagement and silent attrition by analyzing workforce behavior, performance, and engagement data.

Tools and Technologies: Python, Pandas, HR Analytics, Machine Learning, NLP, Data Visualization Tools.

Outcomes: Learn to detect attrition risks, build predictive HR models, and support data-driven talent decisions.





Viral Content Intelligence System

Description: Analyze social media content to identify viral patterns and predict high-performing posts using engagement and trend signals.

Tools and Technologies: Python, NLP, Social Media APIs, Machine Learning, Power BI / Tableau.

Outcomes: Understand content virality factors, build trend-detection models, and optimize content strategies.

Live Industry-Based Projects

Business Risk Probability Simulator

Description: Simulate business risk scenarios by analyzing market, operational, and financial data to estimate risk probabilities.

Tools and Technologies: Python, Statistical Modeling, Monte Carlo Simulation, Machine Learning, Data Visualization.

Outcomes: Learn probabilistic modeling, risk assessment techniques, and scenario-based business forecasting.





Prompt Engineering Studio for Business Automation

Description: Design and test optimized prompts to automate business workflows such as reporting, customer support, and content generation using LLMs.

Tools and Technologies: OpenAI API, LangChain, Prompt Engineering Techniques, Python, LLMs.

Outcomes: Master prompt optimization, build reusable prompt templates, and automate real-world business tasks.

Live Industry-Based Projects

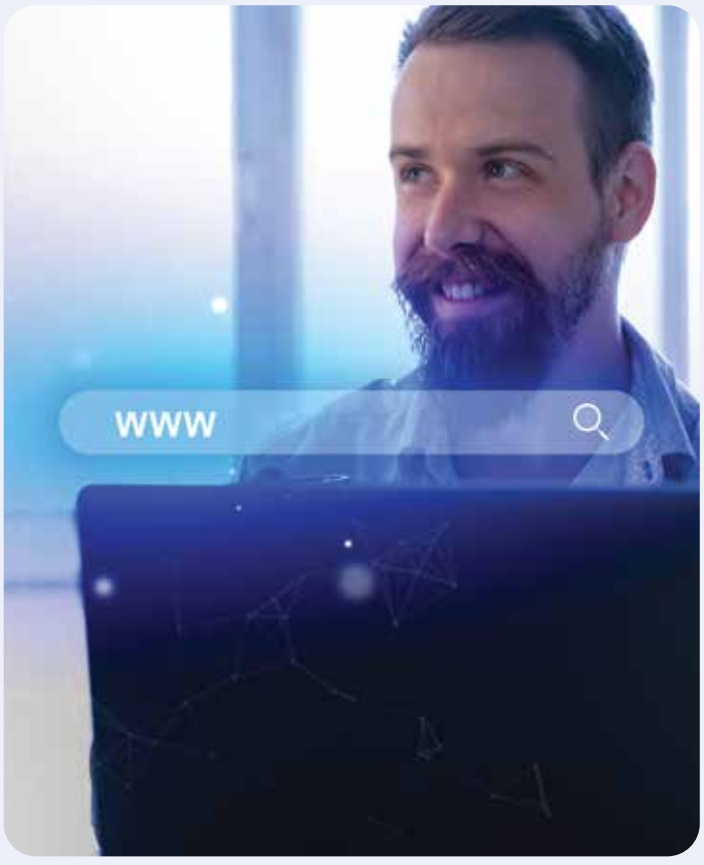
AI Code Assistant for Data Analysts

Description: Build an AI-powered assistant that helps data analysts write, debug, and optimize SQL and Python code.

Tools and Technologies: LLMs, Python, SQL, LangChain, Vector Databases, Prompt Engineering.

Outcomes: Learn to create intelligent coding assistants and improve analytics productivity using AI.





Domain-Specific LLM Fine-Tuning System

Description: Fine-tune large language models on domain-specific datasets to improve accuracy and relevance for specialized use cases.

Tools and Technologies: Hugging Face, PyTorch, Transformers, Fine-Tuning Techniques, GPUs.

Outcomes: Gain hands-on experience in LLM fine-tuning, dataset preparation, and domain adaptation.

Live Industry-Based Projects

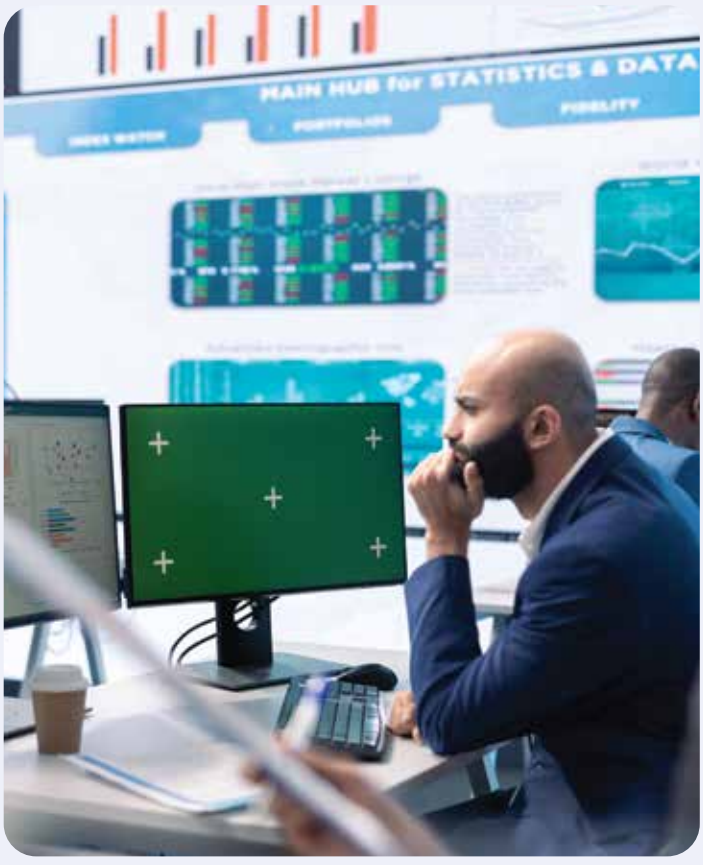
Autonomous Research Paper Summarization LLM

Description: Develop an autonomous system that reads, understands, and summarizes research papers into concise insights.

Tools and Technologies: NLP, Transformers, LLMs, PDF Parsing, Python, LangChain.

Outcomes: Learn document understanding, abstractive summarization, and building research automation tools.





LLM-Powered Contract Risk Analyzer

Description: Analyze legal contracts using LLMs to detect risk clauses, compliance issues, and potential liabilities.

Tools and Technologies: NLP, LLMs, Legal Text Analysis, Python, Vector Databases.

Outcomes: Understand legal document intelligence, risk extraction, and AI-driven contract analysis.

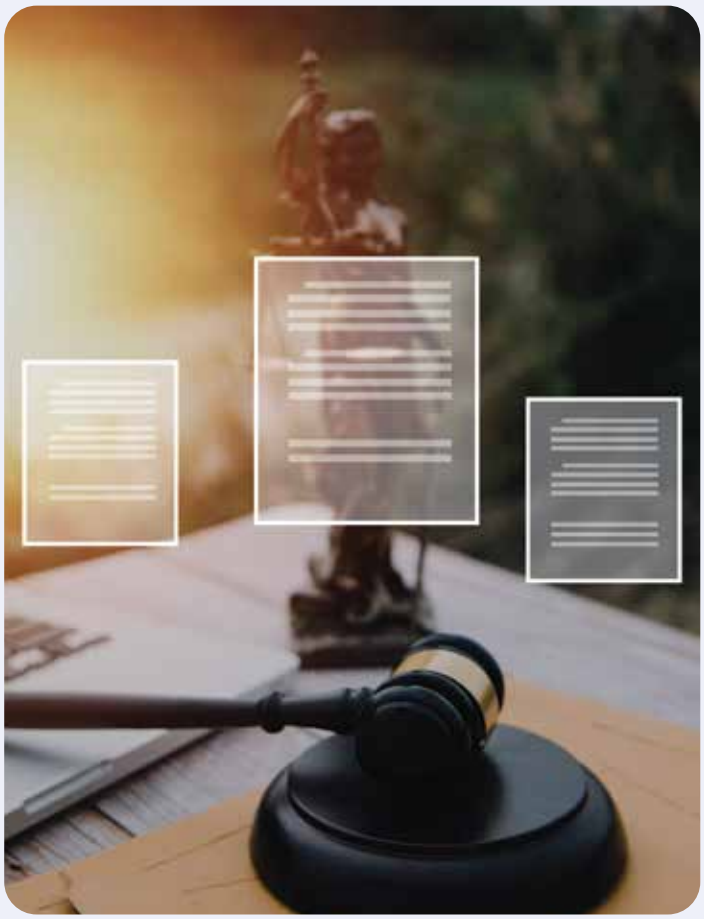
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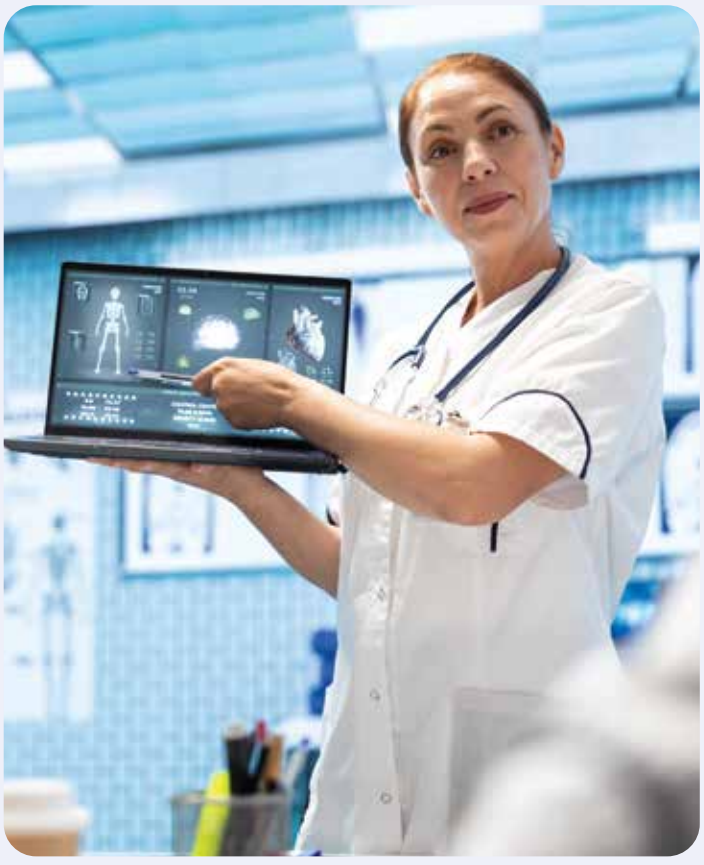
Legal Judgment Search & Reasoning System

Description: Build an AI system that searches, analyzes, and reasons over legal judgments to provide relevant case insights and legal precedents.

Tools and Technologies: NLP, LLMs, Python, Legal Text Processing, Vector Databases, Transformers.

Outcomes: Learn legal document retrieval, semantic search, and AI-based legal reasoning techniques.





Medical Multimodal Diagnosis Assistant

Description: Develop an AI assistant that combines medical images, clinical notes, and patient data to support accurate diagnosis.

Tools and Technologies: Computer Vision, NLP, Multimodal Models, PyTorch, Medical Imaging, LLMs.

Outcomes: Gain experience in multimodal AI, medical data processing, and clinical decision support systems.

Live Industry-Based Projects

Autonomous Business Research AI Agent

Description: Create an autonomous AI agent that conducts market and competitor research by gathering, analyzing, and summarizing business data.

Tools and Technologies: LLMs, LangChain, Web Data Extraction, Python, Vector Databases.

Outcomes: Learn to build autonomous agents for market intelligence and strategic research.





AI Trading & Portfolio Monitoring Agent

Description: Design an AI-driven system to analyze market data, track portfolios, and generate trading insights using predictive models.

Tools and Technologies: Time-Series Analysis, Machine Learning, Python, Financial APIs, Data Visualization.

Outcomes: Understand algorithmic trading basics, portfolio analytics, and real-time monitoring systems.

Live Industry-Based Projects

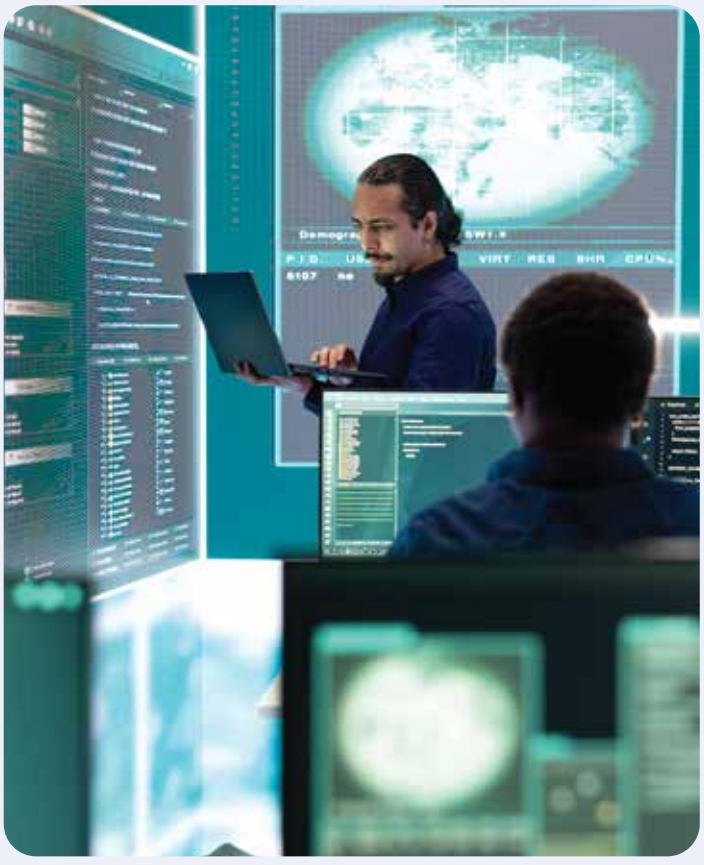
AI Explainability Dashboard for Executives

Description: Build a dashboard that explains AI model decisions in a clear, business-friendly manner for executive stakeholders.

Tools and Technologies: Explainable AI (SHAP, LIME), Python, Dash / Power BI, Machine Learning Models.

Outcomes: Learn to interpret AI models, communicate insights, and build trust in AI systems.





Responsible AI Deployment Framework

Description: Develop a framework to ensure ethical, fair, and compliant deployment of AI systems across business applications.

Tools and Technologies: AI Ethics, Bias Detection, Model Governance, Python, Compliance Tools.

Outcomes: Gain knowledge of responsible AI principles, risk mitigation, and governance best practices.

Live Industry-Based Projects

AI-Powered Drug Discovery Intelligence System

Description: Apply AI to analyze biological and chemical data to accelerate drug discovery and candidate identification.

Tools and Technologies: Deep Learning, Bioinformatics, Molecular Modeling, PyTorch, Graph Neural Networks.

Outcomes: Learn AI-driven drug discovery workflows and biomedical data analysis techniques.





AI-Driven Smart Manufacturing Optimization (Industry 4.0)

Description: Build AI solutions to optimize production processes, reduce downtime, and improve efficiency in smart factories.

Tools and Technologies: IoT Analytics, Machine Learning, Predictive Maintenance, Python, Industry 4.0 Systems.

Outcomes: Gain experience in industrial AI, predictive optimization, and smart manufacturing systems.

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WhatsApp/Call

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